

Question	Answer	Marks
9(a)	number of nuclear disintegrations per unit time	B1
9(b)	activity is proportional to the number of undecayed nuclei	B1
	activity = $(-)$ rate of change of number of undecayed nuclei	B1
	N is proportional to the rate of change of N (so exponential variation)	B1
9(c)(i)	$120 = 180 \exp(-\lambda \times 8.4)$	C1
	$\lambda = 0.048 \text{ min}^{-1}$	A1
9(c)(ii)	half-life = $\ln 2 / 0.048$ = 14 min	A1
9(c)(iii)	line with negative gradient throughout, starting at (0, 180)	B1
	curve with negative gradient passing through (8.4, 120)	B1
	curve with decreasing negative gradient, from $t = 0$ to $t = 24$ min, passing through (14, 90)	B1

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